DOI: https://doi.org/10.29367/issn.1809-3957.13.2018.155.58

Revista SODEBRAS – Volume 13
N° 155 – NOVEMBRO/ 2018

CONTROL AND MONITORING SYSTEM OF ELEVATORS USING INDUSTRIAL PLC SYSTEM

TIAGO TARGINO[1]; ISAN NASSIFFE[2]; LUTER FERRAZ[3]; LEIZER SCHNITMAN[4]; MARCIO FONTANA[5]
1; 2; 3; 4; 5 - UFBA - UNIVERSIDADE FEDERAL DA BAHIA - LABORATÓRIO DE AUTOMAÇÃO INDUSTRIAL, CENTRO DE CAPACITAÇÃO TECNOLÓGICA EM AUTOMAÇÃO INDUSTRIAL
tiago.targino@ufba.br; isanmn@gmail.com; luterferraz@gmail.com; leizer@ufba.br; mfontana@ufba.br

Abstract - This paper presents the development of a control and monitoring system for elevators in a building based on Programmable Logic Controller (PLC). A SCADA platform is developed and a supervisory system could be successfully built. The results of the programmable PLC technology solutions in one hundred and three-story elevator control are discussed.

Keywords: Industrial Automation. SCADA Platform. Elevators. PLC.

I. INTRODUCTION

An elevator is a type of vertical transport equipment that moves people or objects between floors of a building with efficiency and comfortable service. Furthermore, it provides accessibility for people with disabilities, allowing the right to come and go between floors. In modern time, due to rapid population growth in the urban areas, in multi-stored buildings need for elevators is being increased. Due to the vertical expansion of the buildings, the use of two or more elevators working at the same time was usually necessary.

Programmable logic controllers (PLCs) have been widely used for different control applications (ALPHONSUS AND ABDULLAH, 2016) and many academic studies have used small PLCs for elevator control systems, such as: SIEMENS LOGO! (SINGH *et al*., 2013), SIEMENS PLC S7–200 (YANG, ZHU and XU, 2008; SEHGAL and ACHARYA, 2014; CHEN, YIN AND LIU, 2018), EATON 512-DCRC PLC (BERNARD *et al*., 2015) and OMRON CJ1M-PLC (WEI and WANG, 2010). The development of a monitoring system is a methodical approach to problem solving related to the real world. Furthermore, it can avoid incidents, improve the quality, bring safety and increase the reliability of the system. Also, the process of development brings different concepts of Engineering which make a valuable tool for engineering education.

This paper presents a project based on Micro850® PLC from Rockwell Automation to control and monitoring in real time the competition among three elevators in a building with 103 floors using a Supervisory Control and Data Acquisition SCADA system, gathering data on the process and sending commands to the connected devices using a simulated environment. Any of three elevators should be able to pick a passenger, however, the decision of which will do it should be a result of a competition among the elevators and the efficiency is considered to support the decision.

II. HEURISTIC APPROACH

The growing number of tall buildings leads the need of multiple elevators. However, there is a problem related to the control of the elevators usually called Elevator Dispatching Problem (EDP). The elevator dispatching has many control constraints and when not applied correctly it may lead to inefficiency and waste of time. In order to optimize the EDP many algorithms have been developed and some of them are discussed next.

The collective control is the most basic algorithm where the elevator cars always stop at the nearest call in their running direction. It is possible to apply different configurations to this algorithm. For instance, the Collective-up/Selective-down mode where it collects the hall-calls during the up-trip, however, during the down-trip the lowest hall-call is chosen to be visited first. Hence, this operation mode is suitable when the up-direction traffic is intense (MUNOZ *et al.*, 2006).

The Estimated Time to Dispatch (ETD) is an algorithm widely used by manufacturers. Based on the user destination and current floor, the algorithm allocates the best elevator car to the user considering which elevator will transport the user with shortest time. This method not only considers the proximity of the calls but also ponders the direction and number of stops the elevator is going to make before it reaches the desired floor (LATIF, KHESHAIM AND KUNDU, 2016).

Many papers approach the use of fuzzy logic to control a group of elevators (KIM *et al.*, 1998; GUDWIN, GOMIDE AND NETTO, 1998). Considering different parameters, the fuzzy logic is able to determine which elevator should be allocated to the user. Each elevator is considered as a multi-input and single output (MISO) system and the design criteria can be stipulated to minimize different objectives such as: the waiting time and crowding in an elevator (KHIANG, KHALID AND YUSOF, 1995).

This paper applied a modified version of the ETD. Instead of considering the whole-time travel, our algorithm aims to minimize the waiting time considering the direction and number of stops as same as ETD.

III. EXPERIMENTAL SETUP

3.1 – Software

The PLC was programmed using the Connect Component Workbench (CCW) Software provided by Rockwell Automation. The CCW offers three logics of programming: Ladder; Functional Block Diagram and Structured Text. The last one was chosen to realize the project. The monitoring is realized utilizing the Factory Talk View Studio from Rockwell Automation which is a software for developing and running Human-Machine Interface (HMI) applications, designed for monitoring and controlled automated processes and machines (ROCKWELL, 2018). The communications between multiple Rockwell Software products are made utilizing the RSLinx through an Object Linking and Embedding for Process Control (OPC) server (ROCKWELL, 2018). Figure 1 shows the connection between the Rockwell products.

Figure 1 – Connection between Rockwell Automation's business solutions

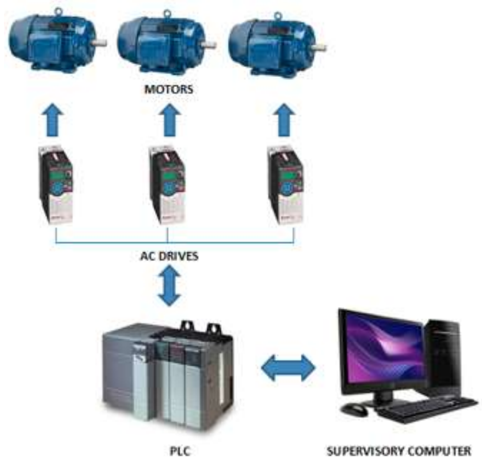
Source: Authors, 2018.

3.2 – Hardware

It was used an expandable Micro850® PLC which uses EtherNet/IP protocol and function as a remote terminal unit (RTU) for SCADA applications with support for Modbus over serial and Ethernet communications (ROCKWELL, 2015). Three PowerFlex 525 AC drives from Allen-Bradley have been used to control the motors. The drives offer embedded EtherNet/IP communications and USB programming. Also, PowerFlex 525 offers a wide range of motor control including: Volts per hertz (V/Hz); Sensorless Vector Control (SVC), Closed loop speed vector control and Permanent Magnet motor control (ROCKWELL, 2016). A set of three-phase induction motors with two poles from WEG have been used to represent the elevators (WEG, 2016). They have rated power of 0.12kW, nominal voltage and current of 220/380V and 1.28/0.741A respectively. The nominal speed is equal to 3430 RPM and nominal frequency 60Hz (WEG, 2016).

Also, a single supervisory computer is used. It gathers the data on process and sends control commands to the field connected devices. Figure 2 shows the connection diagram englobing all the components of an experimental setup.

Figure 2 – Components of experimental setup



Source: Authors, 2018.

IV. PLC LOGIC STRUCTURE

Before the beginning of the project, a few requirements were made and certain operation rules were decided. The elevator must move at a specific speed for distances below 20 floors and at another speed for distances equal or above 20 floors. The acceleration time must be equivalent to 2.5% of the travel time. The elevator must stop within a reasonable margin of error (0,1%) of the expected position in each floor. The elevator must stop and take in consideration the opening and closing time of the door. Finally, to simulate the elevator and its system, a PLC will be used as the system, controlling three motors through AC drives. The system will be simulated through a SCADA platform.

The programming of the elevator system was divided into 3 sections: (1) when somebody calls the elevator from outside the system needs to determine which elevator should pick the person up based on the floor called and the desired direction; (2) when someone from inside the elevator wants to go to a specific floor, the system also needs to be able to process the command; and (3) it is structured how the elevator should operate, its speed, when to stop, where to go, open the door, close the doors and functionalities specific system of the PLC, fulfilling its tasks and requirements.

4.1 – Calling from outside the elevator

Each elevator has its own queue of floors where it needs to go and the direction to where the person that called wants to go. Thus, each elevator calculates based on its own queue the time it will take to reach that call, the one with the lowest time is the selected one to respond to it.

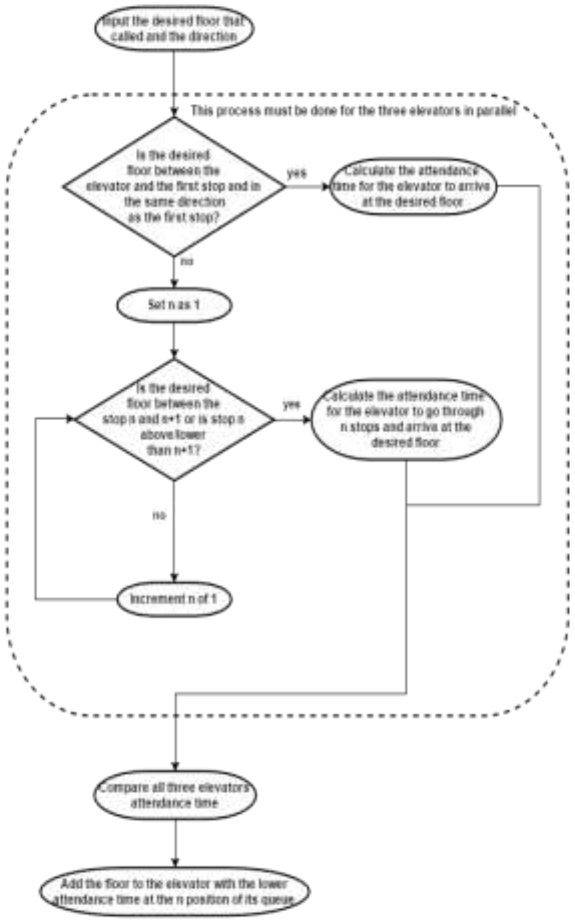
The time calculation takes in consideration two possibilities, if the elevator queue is empty, then all that it needs to do is calculate the time taken from where it is to where it will go based on its medium speed. If there is a queue, it will check if the floor to where it was called is between its position and the floor to where it is going and if both have the same direction. When the called floor is not between or the direction is not the same, then the time between where it currently is and the first floor on the queue using the medium speed and considering the opening and closing door time. After this then it does the same thing for the floor in the second of the queue and so

ISSN 1809-3957

on until identifying where on the queue the called floor should be allocated.

After the attendance time of all three elevators are estimated, the called floor is assigned to the elevator with the lowest attendance time. The flowchart presented in Figure 3 illustrates how the decision is made.

Figure 3 – Flowchart of calling from outside the elevator



Source: Authors, 2018.

4.2 – Calling from inside the elevator

This situation is simpler and based on the situation calling from outside the elevator. As the command is issued from inside the elevator, there is no choosing which elevator will be assigned the task, because it is already decided by the fact that the person is already inside one elevator. The challenge of this task is to identify where to allocate the desired floor in the elevator queue. The data utilized in this process is the actual position of the elevator, the direction in which the elevator is moving and to which floor the person wants to go.

In case the queue is empty, the desired floor is allocated as first. If the elevator is in motion then two conditions are monitored, the first is if the desired floor is between the floor in the position n-1 and n in the queue. The second condition is if the floor in the queue position n is higher than n-1, in the desired floor is above the elevator. In case the desired floor is below the elevator than it checks if the assigned floor in the queue as n is lower than n-1. If any of these two conditions are true then the floor is allocated before n. Figure 4 shows a flowchart with the calling from inside the elevator.

4.3 – Elevator Functions

The elevator structure has the objective of controlling the motor as necessary. Each elevator has its own elevator block, which monitors the queue of floors to where it needs to go. When a floor is assigned to the elevator, firstly is verified if the floor as above its current position or below and then the flag of its direction is set. Also, is verified the distance between the destination and starting point to determine the maximum speed the elevator will travel. Finally, based on the time of travel calculated through the medium speed of the elevator, the time for acceleration from zero to the maximum speed is calculated. After all this is done, the motor is started with rotation direction related to the elevator direction, pre-determined time for motor acceleration and maximum speed.

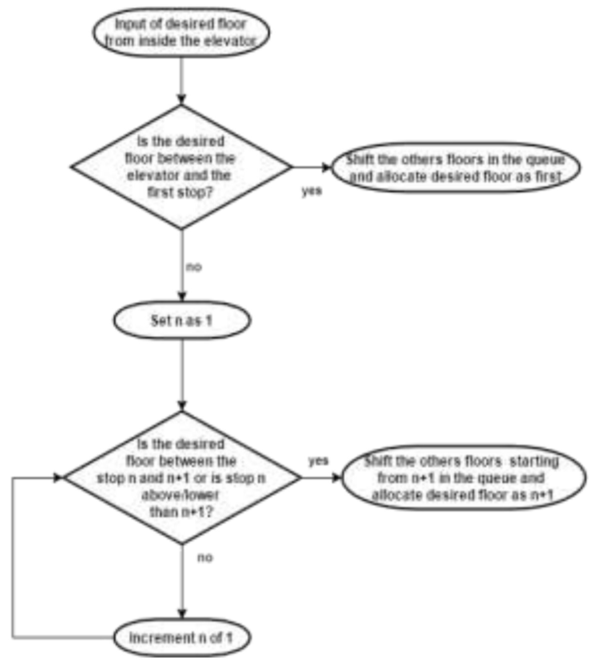
While the motor is operating, the elevator keeps updating its current position and the distance between the first floor on the queue. When a certain threshold for the distance is reached the stop protocol begins. Figure. 5 shows the flowchart referent to the elevator functions.

The stop protocol consists of reducing the motor speed based on ta simple proportional error controller with a constant value as shown in equation (1) to guarantee a minimum speed to achieve a soft stop and precision in its position before opening the door. Where K_p is the proportional gain, E is the instantaneous error and C is the controller output with zero error. The values of K_p and C used were determined through trial and error, obtaining less than 2cm of error for every stop made.

$$V = K_p \times E + C \tag{1}$$

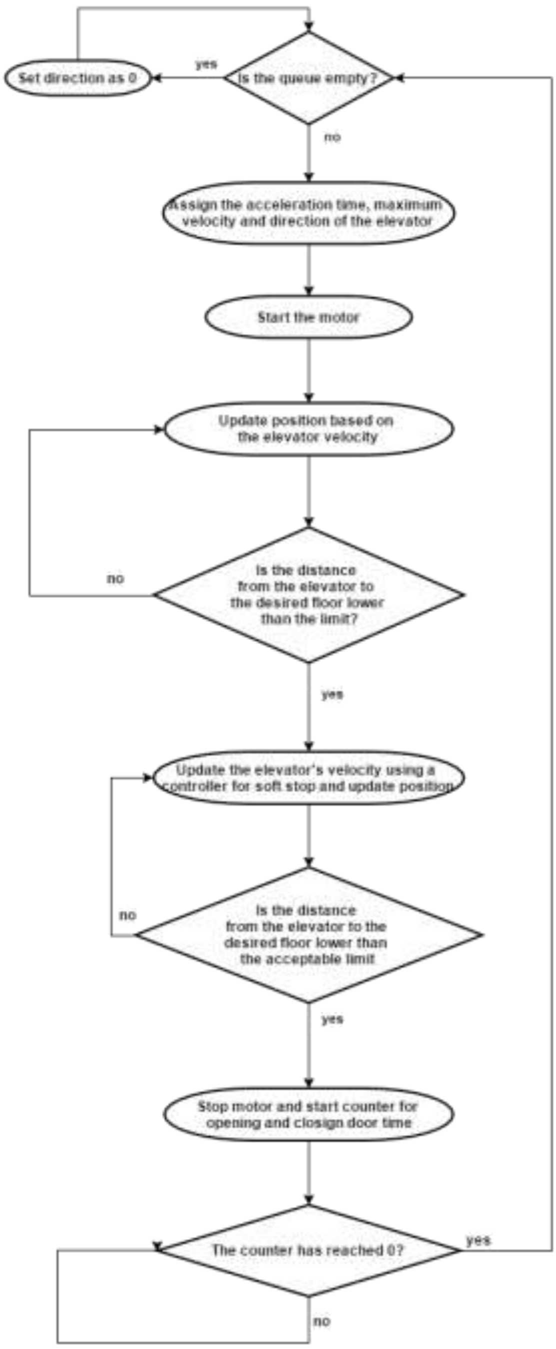
After stopping on the desired floor, the motor is put on hold until the opening and closing time of the door is past and the present floor is removed from the queue. If the queue is empty the direction of the elevator is set to zero to indicate no movement, otherwise the whole process is repeated until the queue is empty.

Figure 4 – Flowchart of calling from inside the elevator



Source: Authors, 2018.

Figure 5 – Flowchart of elevator functions



Source: Authors, 2018.

V. RESULTS

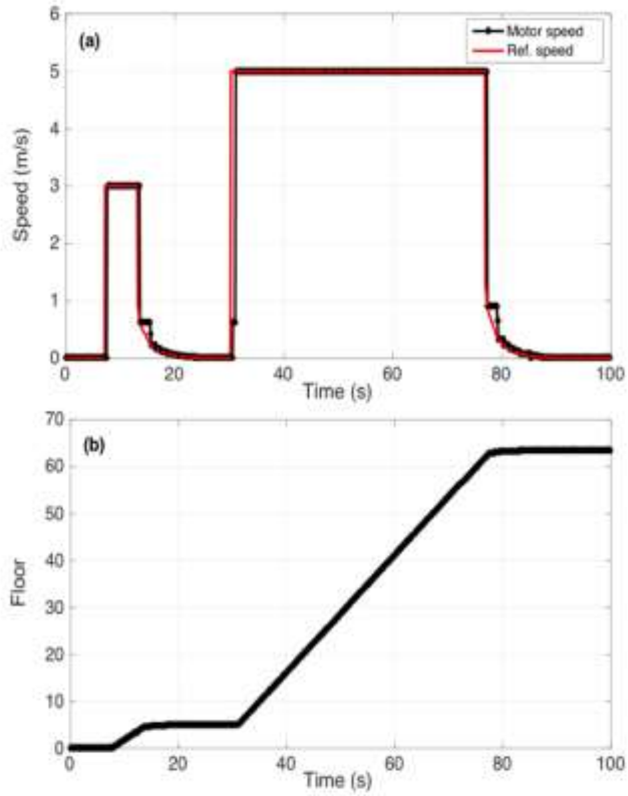
5.1 – PLC Control

Figures 6-a and 6-b, respectively, show the motor speed and the current floor of the elevator. The following parameters were set:

- Maximum speed below 20 floors: 3m/s
- Maximum speed above 20 floors: 5m/s
- Time for opening and closing doors: 6s

Two tests were conducted to verify the PLC control speed. Firstly, the elevator was positioned in the lobby (0 floor), then it was sent to the 5th floor. In this case, the elevator should move with maximum speed 3m/s because it is moving within 20 floors. Afterwards, the elevator was sent to the 65th floor. In this case, the elevator should have maximum speed of 5m/s because it is moving more than 20 floors.

Figure 6 – PLC control Status: (a) motor speed and (b) current floor of the elevator



Source: Authors, 2018.

We observed that the elevator reaches the speed limit and does not go over it (3m/s, 5m/s) in both cases and the breaking control works appropriately.

5.2 – Results of Field Trials

The first test was disabling the other two elevators to check if the allocating queue system was working. The floors called are represented on the Calling Order column while the resulting queue from this test is represented by the column Elevator "X" Queue. The Elevator 1 started at the first floor and its result is shown on Table 1.

Table 1 – First queue test

Elevator 1 Queue		**Calling Order**	
Floor Called	**Direction Called**	**Floor Called**	**Direction Called**
12	Up	12	Up
25	Up	20	Down
60	Up	25	Up
81	Down	60	Up
20	Down	81	Down

As expected, the elevator queue put all floor called with the same direction as the first call with higher priority and taking into account their disposition. As for the calls with the opposite calling direction, they were allocated at the decrescent order, as the direction is downward, thus the system queue allocation worked as planned.

For the second test, all elevators were available and all started from the first floor. The results are on Table 2 in the same format as the former.

ISSN 1809-3957

Table 2 - Second queue test

Elevator 1 Queue		Calling Order	
Floor Called	**Direction Called**	**Floor Called**	**Direction Called**
10	Up	10	Up
Elevator 2 Queue		12	Up
Floor Called	**Direction Called**	5	Down
12	Up		
Elevator 3 Queue			
Floor Called	**Direction Called**		
5	Down		

Now the focus of the task was to verify if the floor distribution among the elevators were correct. As expected, the second calling was attributed to the Elevator 2 because of the time Elevator 1 had to stay waiting for the closing and opening of the doors. As the calling direction was different, the Floor 5 went to the Elevator 3, because it had to be the last call in all elevators queue, so the Elevator 3 that was empty were the one with lower time.

The last test, Elevator 1 started at the fiftieth floor, Elevator 2 at twenty-fifth floor and Elevator 3 at the first floor. The calling order and consequent queue are presented at Table 3.

This third test assured the results of test 2 as the elevators followed the determined principles of where on the queue it would be allocated, considering the direction of the first call, and the distance between the elevator and the call, including opening and closing time.

Analyzing all results (Table 1, Table 2, and Table 3) was considered satisfactory and reached our expectations and demands as the calling distribution seemed efficient

Table 3 – Third queue test

Elevator 1 Queue		Calling Order	
Floor Called	**Direction Called**	**Floor Called**	**Direction Called**
70	Up	10	Up
80	Up	80	Up
Elevator 2 Queue		60	Down
Floor Called	**Direction Called**	15	Up
60	Down	30	Up
Elevator 3 Queue		70	Up
Floor Called	**Direction Called**		
10	Up		
15	Up		
30	Up		

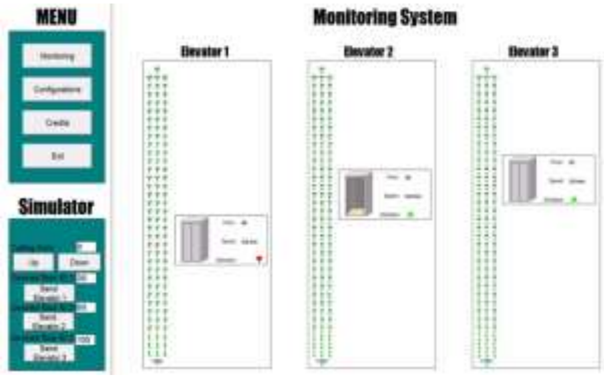
.

5.3 – Human Machine Interface

For monitoring and simulation purposes, a simple GUI was designed, as it is illustrated in Figure 7. With utility in mind, almost all aspects of monitoring and control aspects fit in one screen. Also, there is an animation for the opening and closing doors time (Figure 8).

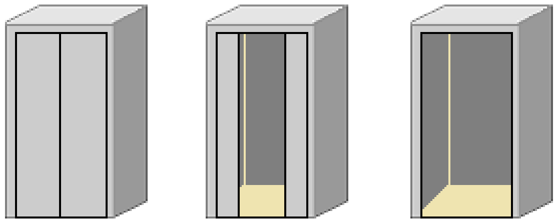
Also, it was implemented a monitoring section, in which the most important attributes of the elevator system are described, such as its speed, acceleration, position and the current state of the automatic doors. It is possible to observe that in the main screen, it was also included a section for simulation purposes, in which the user inputs current and the desired floor.

Figure 7 – Human-Machine Interface: monitoring screen



Source: Authors, 2018.

Figure 8 – Sequence animation



Source: Authors, 2018.

VI. CONCLUSION

Elevator control systems are a common problem in both residential and industrial settings. Thus, developing an efficient algorithm, as well as an easy to use and functional interface in conjunction with a PLC based system proved to be an enlightening educational endeavor. In this study, real time controlling and monitoring of a one hundred and three-story elevators system based on PLC is realized. The system developed can be used either in industrial automation or simulation educational purposes. Control and monitoring systems of elevators using respectively PLC and SCADA is an important issue of Electrical Engineering Education and it is becoming a requirement for the job market. During our study, it was verified that smalls PLC are powerful enough devices to satisfy both academic and professional purposes.

VII. ACKNOWLEDGMENT

This study was supported by CTAI – "Centro de Capacitação e Tecnológica em Automação Industrial" at the Federal University of Bahia (UFBA) and Rockwell Automation do Brasil Ltda

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IX. COPYRIGHT

Submetido em: 19/06/2018
Aprovado em: 21/09/2018